LOCOMOTIVE TRACTION MOTOR ROTOR LOCK REDESIGN

Wabtec CORPORATION

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BACKGROUND

Traction motors convert electrical energy into mechanical force: These motors power locomotives transporting billions of tons of freight annually.

Wabtec remanufactures tens of thousands of motors each year: This restores performance while reducing costs & environmental impact.

Bearings are at risk during shipment: Vibrations and micro-movements can cause false brinelling, a type of wear that compromises motor quality and lifespan.

Existing rotor locks offer partial protection: Bearings near the air intake are now better protected under radial shock loads, resulting in a more robust design that enhances protection during transport.

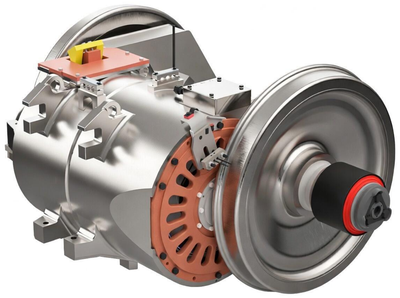


Figure 1: Traction Motor Assembly

OBJECTIVE

Redesign a rotor lock to **protect all internal bearings** in locomotive traction motors during shipment, **ensuring reliable performance**, **minimizing maintenance**, and **extending the service life** of Wabtec's remanufactured components.

KEY REQUIREMENTS

- ✓ Rotor lock shall accommodate **EA & B** motor types.
- ✗ Device shall withstand **over 10,000 lb** axial and radial shock loads.
- ✓ Rotor lock assembly shall adhere to OSHA single individual lift standards **(< 50 lb)**.
- ✓ Device shall be secured/installed in **<10 minutes**.
- ✗ Device shall meet a minimum **1.5 yielding FOS**.
- Rotor shaft shall rotate **0 degrees** during shipping processes.
- Assembly shall function in temperatures from **-40 °C to 60 °C**.
- Device shall withstand hundreds of thousands of cycles without deformation.
- The cost of one rotor lock assembly shall not exceed **$300** at-scale.

Special thank you to: Sean Cummings, Andy Payne, Behruz Rashidov, Jianliang Xiao, & Design Center Colorado (DCC) Staff.

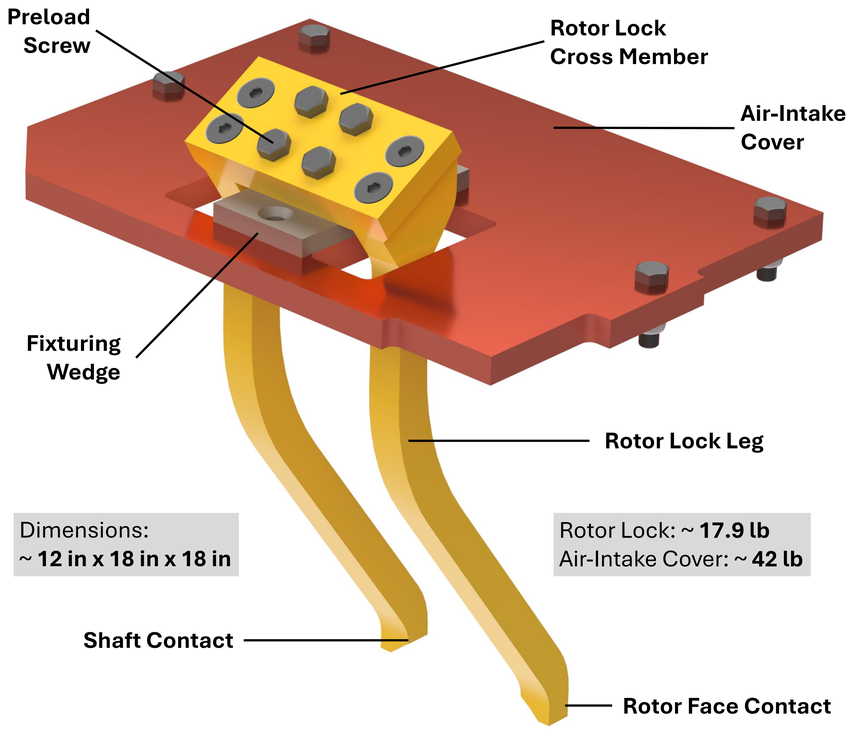


Figure 2: Rotor Lock Assembly Overview

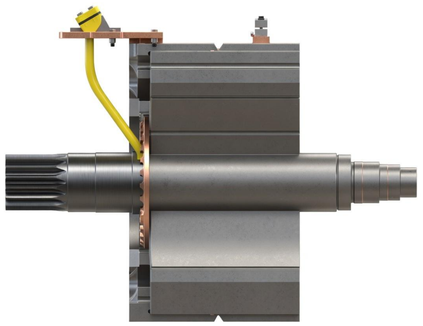


Figure 3: Cross-section Side-view

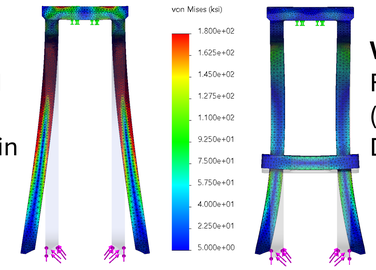
Figure 4: Cross-section Isometric-view

ANALYSIS

A2 Tool Steel is used for its **machinability** and **high-strength** attributes after heat treatment. Simulations were conducted on both component and assembly-level models and supported by manual calculations.

Secondary analysis conducted with outward radial forces on legs.

Without Bracing
Factor of Safety: <1
(Screw Failure)
Displacement: ~ 3 in



With Bracing
Factor of Safety: >1.5
(Anticipated)
Displacement: ~ 0.4 in

Figure 5: Leg Displacement Comparison with Overlay

TESTING

Testing forces scaled for safety and feasibility.
~ 30,000 lb → ~ 10,000 lb; near-linear scaling.

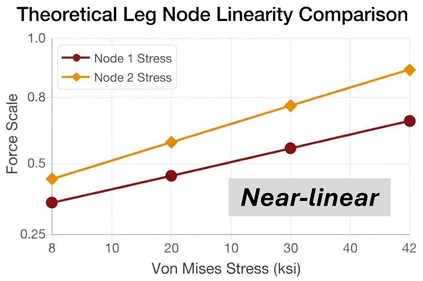


Figure 6: Theoretical Scaling Linearity

COMING SOON

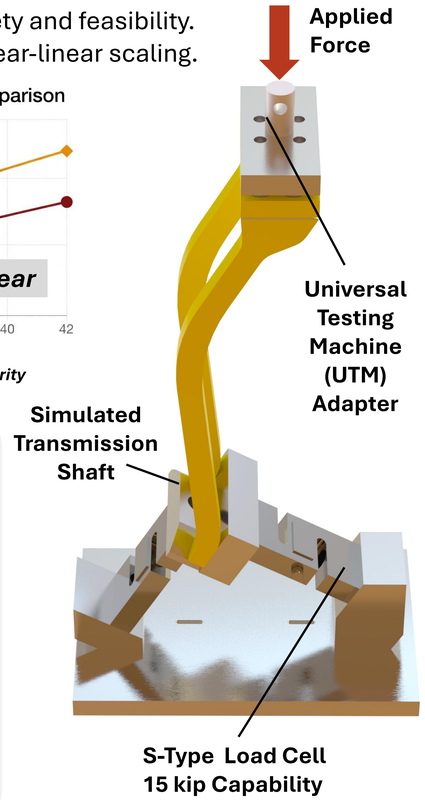


Figure 7: Testing Apparatus Overview

FUTURE WORK

Addition of a lateral bracing component assists in combatting harsh radial loads the device must withstand.

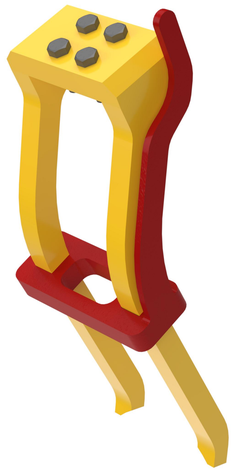


Figure 8: Cross-brace Design

Design Improvements:

Cast Components
- Reduces components
- Manufacturing efficiency
- Lowers production costs

Lateral Bracing
- Resists outward forces
- Improve structural stability
- Removable to bypass air-intake bar
- Locks into place on rotor lock to prevent movement

Ongoing efforts will prioritize the refinement of brace integration.